Exam in Numerical Analysis E3, FMN050, 050113

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This exam lasts 08:00–11:00. In order to pass, a minimum of 15 points out of the total 30 is required. The final grade of the course will be: for **grade 3**, 15–19 points are required; **grade 4**: 20–24 p, **grade 5**: 25–30 p (preliminary figures).

During the exam you are allowed to use a pocket calculator and the course lecture notes (copies of slides). No other material is allowed. Please answer the questions in Swedish or English.

Problems and Solutions

A. Condition and Error analysis

1. Let the function F be defined by $F(x, y) = x \sin y$. Find an approximation to $\delta F = F(x + \delta x, y + \delta y) - F(x, y)$. **(2p)**

2. Suppose that $y \approx \pi/2$. What can you conclude about the relative error $\delta F/F$? **(1p)**

3. What is the condition number for the computation of F at $x = 1$ and $y = \pi/2$? **(2p)**

Solution. Differentiation yields

$$\frac{\mathrm{d}F}{F} = \frac{\sin y \,\mathrm{d}x}{x \sin y} + \frac{x \cos y \,\mathrm{d}y}{x \sin y} = \frac{\mathrm{d}x}{x} + \frac{\mathrm{d}y}{\tan y}.$$

Hence $\delta F/F \approx \delta x/x$. The condition number is 1.

B. Interpolation and Least squares

1. Interpolate the data in Table 1 by a quadratic Newton interpolation polynomial. **(2p)**

x	3	4	6
y	10	4	4

Table 1: Data for problem **B**1.

2. Fit a first degree polynomial $y^*(x)$ to the same data using the least squares method. **(2p)**

3. Compute the deviation between your straight line $y^*(x)$ at the given data points and the original function $y(x)$, and show that $y - y^*$ is orthogonal to y^*. **(2p)**

Solution. The interpolation polynomial is

$$P(x) = c_0 + c_1(x-3) + c_2(x-3)(x-4).$$

Inserting the data we get $c_0 = 10$ and

$$c_0 + c_1 = 4$$

which gives $c_1 = -6$, and finally

$$c_0 + 3c_1 + 6c_2 = 4$$

to get $c_2 = 2$. The polynomial is

$$P(x) = 10 - 6(x-3) + 2(x-3)(x-4).$$

Fitting a straight line means putting $y^*(x) = a(x-4) + b$, which yields the overdetermined system

$$\begin{aligned} -a + b &= 10 \\ b &= 4 \\ 2a + b &= 4 \end{aligned}$$

with normal equations

$$\begin{aligned} 5a + b &= -2 \\ a + 3b &= 18 \end{aligned}$$

which yields $a = -12/7$ and $b = 46/7$. Then $y^*(x) = -12(x-4)/7 + 46/7$ and table 1 is extended to

x	3	4	6
y	10	4	4
y^*	58/7	46/7	22/7
$y - y^*$	12/7	−18/7	6/7

and we see that $58 \cdot 12 - 46 \cdot 18 + 22 \cdot 6 = 0$, proving the orthogonality relation.

C. Integration

1. Compute

$$\int_0^1 x \cos x \, \mathrm{d}x$$

by Romberg's method with an accuracy of five digits. For full credit you should demonstrate in your Richardson extrapolation tableau that you have obtained the requested accuracy. **(5p)**

h	$T(h)$	$\Delta/3$	$S(h)$	$\Delta/15$	$R(h)$
1/2	.354471				
		6838			
1/4	.374984		.381822		
		1698		−3	
1/8	.380078		.381776		.381773

Solution. Romberg's method produces the extrapolation table above. The small change in the last extrapolation step indicates that the requested accuracy has been obtained. Compare also the exact result, $\cos 1 + \sin 1 - 1 \approx 0.381773291$.

D. Newton's method

1. Solve the the nonlinear equation

$$x = 2 - \ln x$$

using Newton's method. Determine all roots to full precision in your calculator. **(3p)**

Solution. Let $f(x) = x + \ln x - 2$ with derivative $f'(x) = 1 + 1/x$. The iteration is

$$x_{n+1} = x_n - \frac{x_n^2 + x_n \ln x_n - 2x_n}{x_n + 1}.$$

By graphing f it's easy to see that there's only one root, near $x = 1.5$. This starting value produces $x_1 = 1.556720935$, $x_2 = 1.557145576$ and $x_3 = 1.557145599$.

E. ODE Initial Value Problem

1. Determine $y(0.4)$ when $y(0) = 1$ and

$$\dot{y} = y,$$

when you solve the problem using Heun's method

$$\begin{aligned}
\dot{Y}_1 &= f(y_n) \\
\dot{Y}_2 &= f(y_n + h \cdot \dot{Y}_1) \\
y_{n+1} &= y_n + h(\dot{Y}_1 + \dot{Y}_2)/2
\end{aligned}$$

with step size $h = 0.4$. **(3p)**

Solution. We obtain

$$\begin{aligned}
\dot{Y}_1 &= y_n \\
\dot{Y}_2 &= y_n + h \cdot \dot{Y}_1 = (1 + h)y_n \\
y_{n+1} &= y_n + h(y_n + (1 + h)y_n)/2 = (1 + h + h^2/2)y_n.
\end{aligned}$$

With the given data, we have

$$y(0.4) \approx (1 + 0.4 + 0.08) \cdot 1 = 1.48.$$

F. Linear systems

1. When solving linear systems $Ax = b$ with iterative methods, the matrix A is split into $A = L + D + U$. The Jacobi method iterates according to

$$Dx^{(k+1)} = -(L+U)x^{(k)} + b$$

Use this method, starting from $x^{(0)} = 0$, to perform one iteration on the system

$$\begin{bmatrix} 5 & -2 & 0 \\ 1 & 4 & -2 \\ 4 & -3 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}.$$

(3p)

Solution Here we have

$$D = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

and

$$L + U = \begin{bmatrix} 0 & -2 & 0 \\ 1 & 0 & -2 \\ 4 & -3 & 0 \end{bmatrix}.$$

Taking one iteration from 0 means that we have to solve

$$D = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 9 \end{bmatrix} x^{(1)} = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$$

so the solution is $(1/5, -1/2, 1/3)$.

G. Quick questions (True or false; 0.5p for each correct answer).

1. In a linear system of equations $Ax = b$ the condition number is $\kappa[A] = \|A\| \cdot \|A^{-1}\|$. **TRUE**

2. In an overdetermined linear system $Ax \approx b$ the least squares method finds an approximate solution x^* which is orthogonal to b. **FALSE**

3. Polynomial interpolation always a *linear* equations problem, independent of the choice of polynomial basis (e.g. monomial basis, Lagrange interpolation, Newton interpolation etc.). **TRUE**

4. If the explicit Euler method is used to solve the initial value problem $\dot{y} = \lambda y$ with $\lambda \in \mathbb{C}$, then $y_n = P(\lambda \cdot \Delta t) y_{n-1}$, where Δt is the time step and P is a polynomial. **TRUE**

5. LU-factorization of a dense $n \times n$ matrix A costs only $\mathrm{O}(n \log n)$ operations to carry out. **FALSE**

6. The fixed-point iteration method for a problem $x = g(x)$ converges only if $\|g'(x^*)\| < 1$. **TRUE**

7. Jacobi's method for solving $Ax = b$ is a fixed-point iteration method. **TRUE**

8. The fast Fourier transform (FFT) is "fast" only when the size n of vector and matrix is a prime number. **FALSE**

9. If $\dot{y} = f(y)$ is solved using the implicit Euler method then one must solve a nonlinear equation of the form $y = hf(y) + \psi$ on each time step. **TRUE**

10. The trapezoidal rule for computing $\int_a^b f(x)\,\mathrm{d}x$ is of order 2, i.e., the numerical result is exact for all polynomials degree 1 or lower. **TRUE**

(5p total)